DC MOTOR SPEED CONTROL - SIMULATION OUTPUTS

Classical controllers : P | PD | PI | PID

EE635T - CSPML in Practice Lab 7 - Classical controllers for DC motor speed control

YOUR NAME HERE | YOUR ROLL NUMBER | generated 2026-10-21

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MOTOR PARAMETERS
  J   rotor inertia             0.0100   kg.m^2
  b   viscous friction          0.1000   N.m.s
  K   = Kt = Ke                 0.0100   N.m/A
  R   armature resistance        1.000   ohm
  L   armature inductance        0.500   H

OPERATING POINT
  reference speed                 1.00   rad/s
  simulation window               3.00   s
  electrical time const L/R     0.5000   s
  mechanical time const J/b     0.1000   s
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CONTROLLER
  C(s) = Kp + Ki/s + Kd.s/(1+Tf.s)
  Kp    100.00      Ki     80.00
  Kd     5.00       Tf     0.000 s
  derivative acts on the measurement

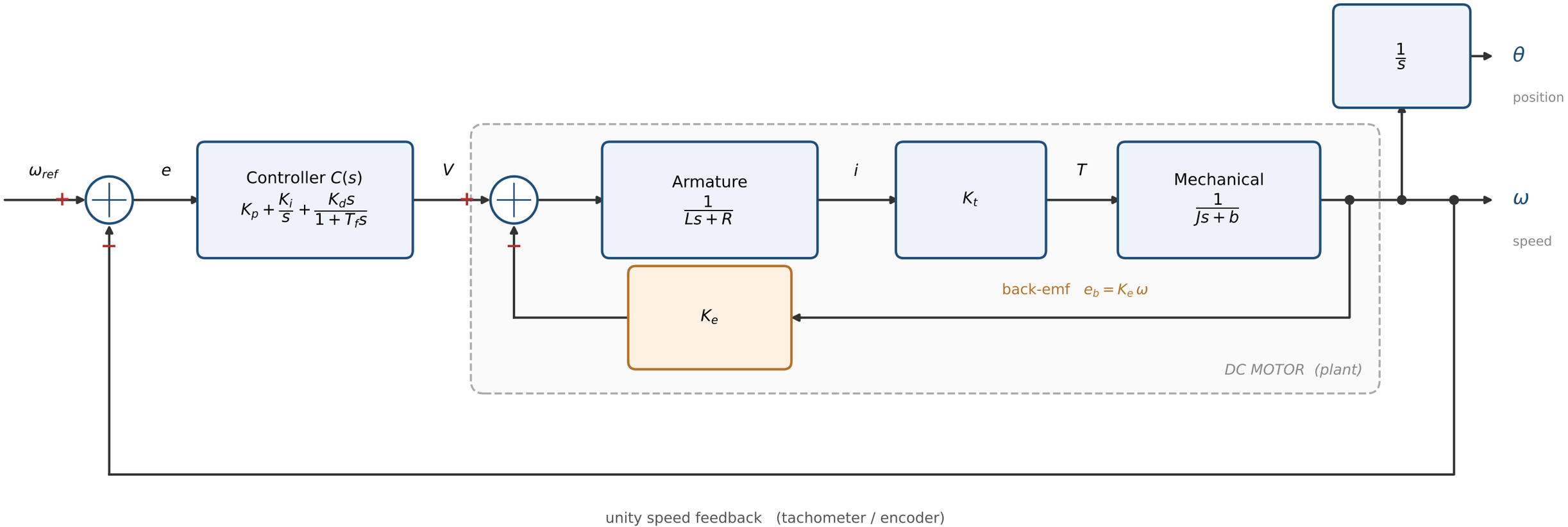
PLANT
  Delta(s) = (Js+b)(Ls+R) + K^2
           = 0.005 s^2 + 0.06 s + 0.1001
  w(s)/V(s)      = K / Delta(s)
  theta(s)/V(s)  = K / [s.Delta(s)]
  i(s)/V(s)      = (Js+b) / Delta(s)
  open-loop poles   -9.997, -2.003
  open-loop DC gain 0.09990 rad/s per volt
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KEY ASSUMPTIONS
  linear time-invariant armature circuit, constant R and L
  constant field flux, so Kt = Ke = K in SI units
  saturation, hysteresis and eddy-current losses neglected
  brush drop, commutation and armature reaction neglected
  continuous-time controller, zero initial conditions
  rigid shaft, constant inertia, no backlash
  purely viscous friction, no Coulomb friction or cogging
  no external load torque, so i_ss = b.w_ss / K
  ideal amplifier and ideal noise-free speed measurement
  step reference applied at t = 0
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HEADLINE RESULT
  controller     tr (s)    ts (s)    Mp (%)    ess (rad/s)
  P             0.0990    0.5672     24.92       0.09099
  PD            0.1521    0.3921      3.10       0.09099
  PI            0.1050    1.0684     21.26       0.00000
  PID           0.1641    0.2471      1.78       0.00000
  most suitable : PID  -  zero steady-state error with the shortest settling time
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CONTENTS 2 block diagram 3 P 4 PD 5 PI 6 PID 7 comparison of all four 8 performance metrics

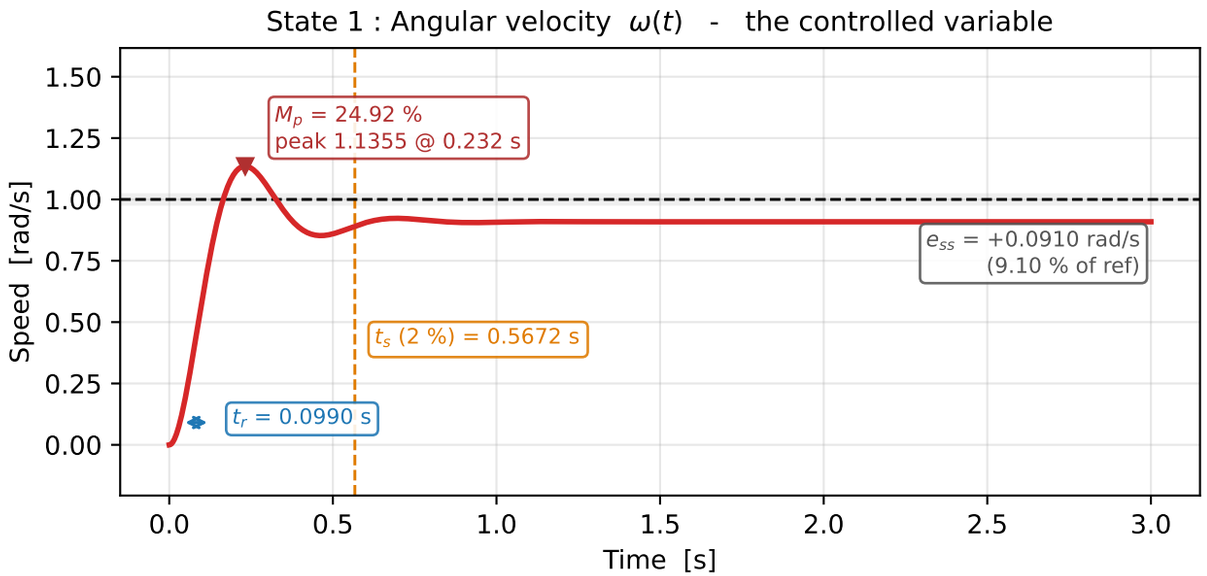
Closed-loop block diagram of the DC motor speed-control system

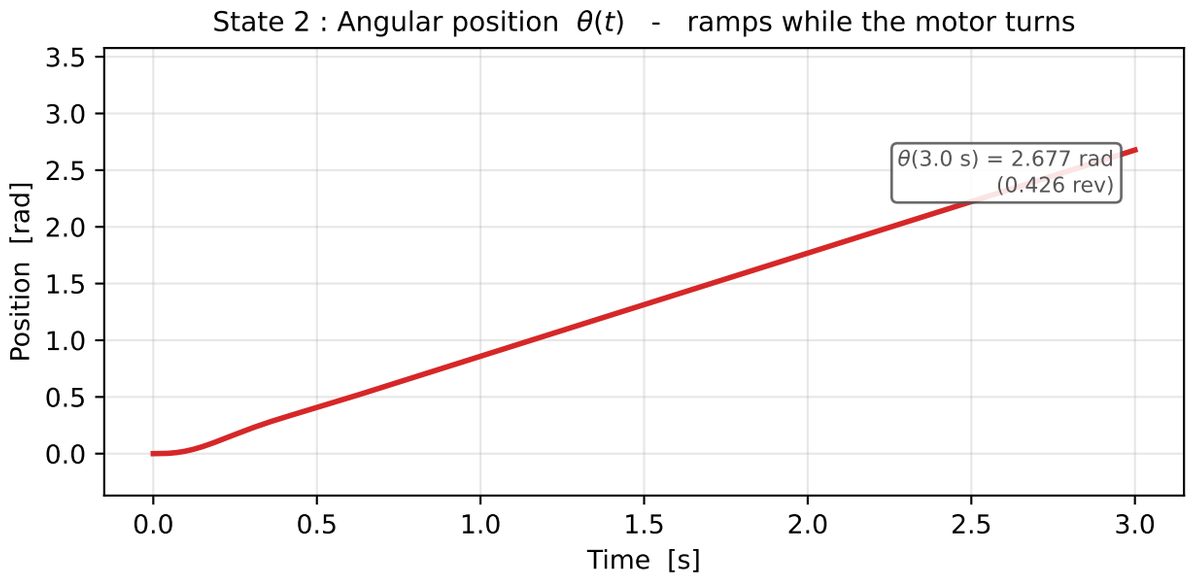


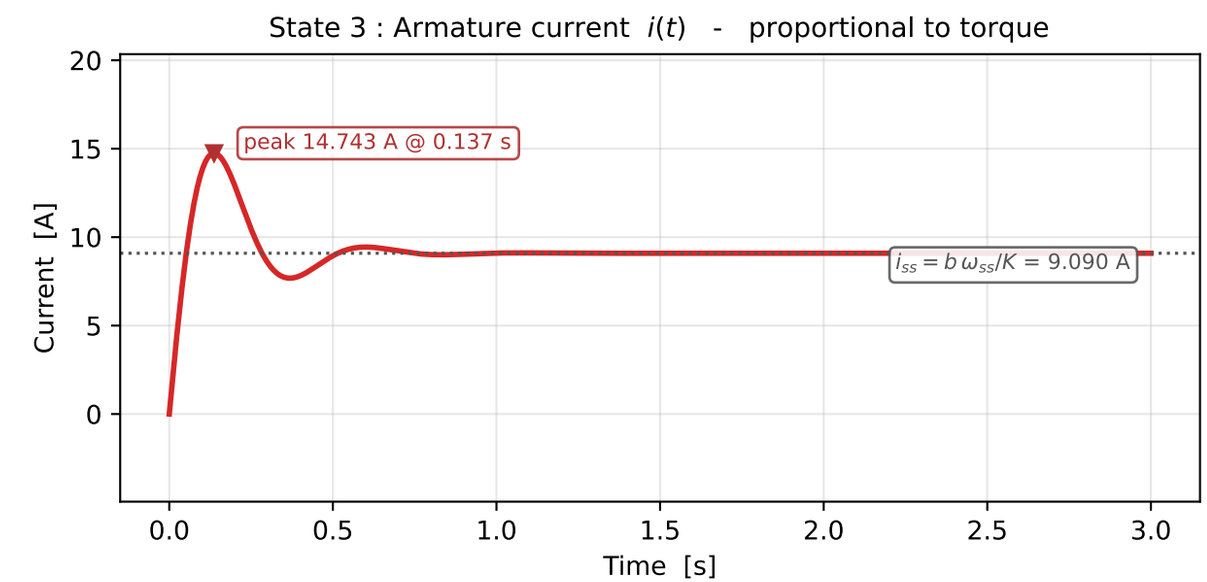
P controller Kp = 100.00 Ki = 0.00 Kd = 0.00 Tf = 0.000 s
motor: J = 0.0100 b = 0.1000 K = 0.0100 R = 1.000 L = 0.500 H derivative on measurement

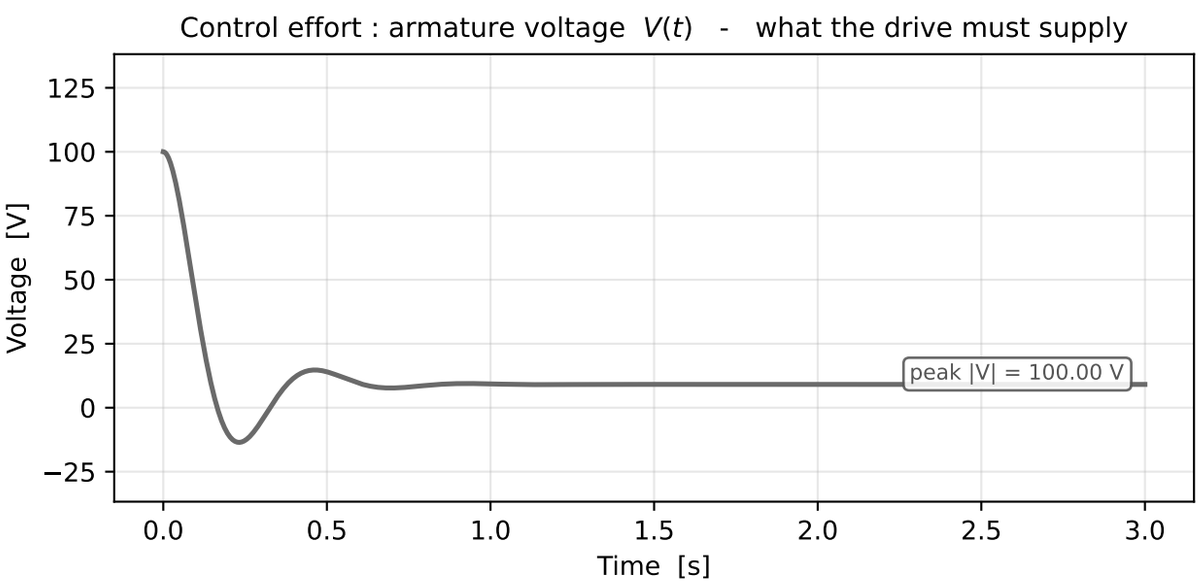
STABLE - every closed-loop pole lies in the left half-plane (rightmost Re = -6.000)

`closed-loop poles:  -6.000+13.565j, -6.000-13.565j`





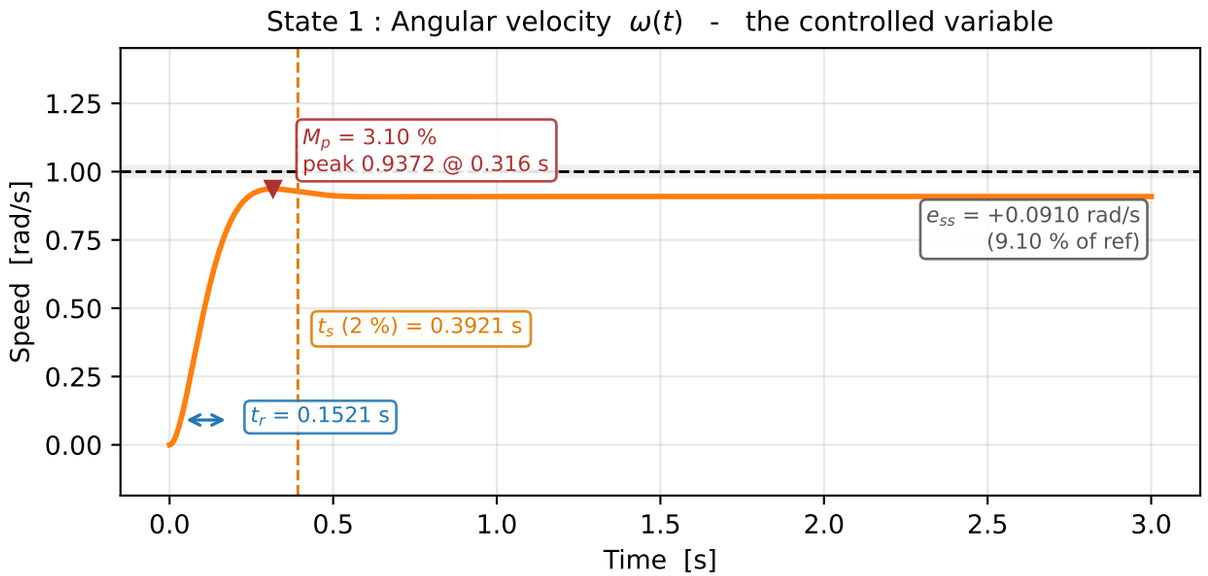


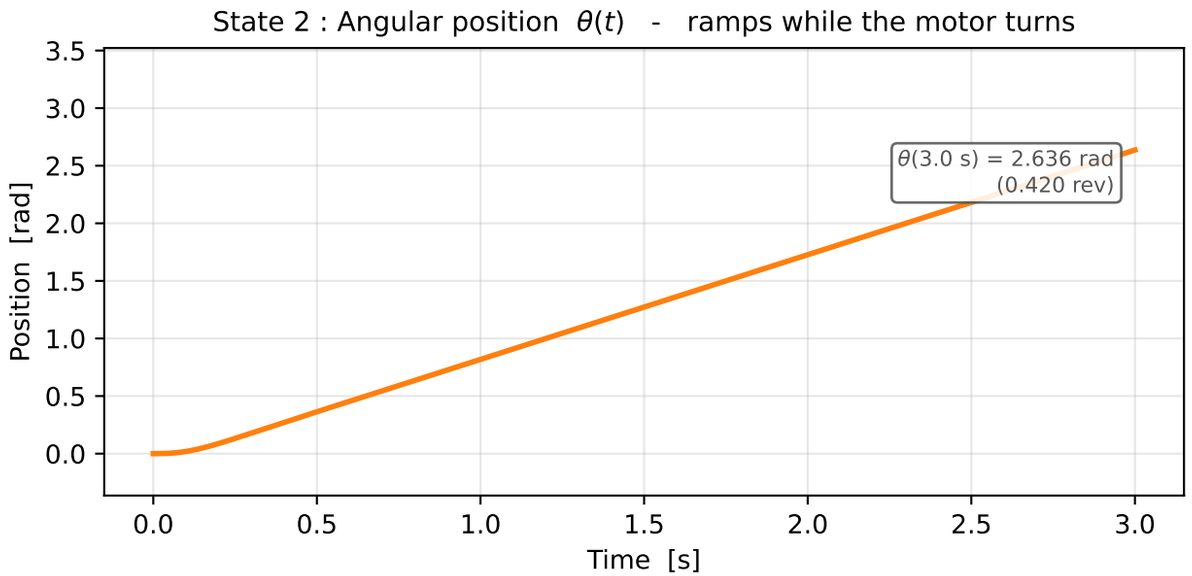


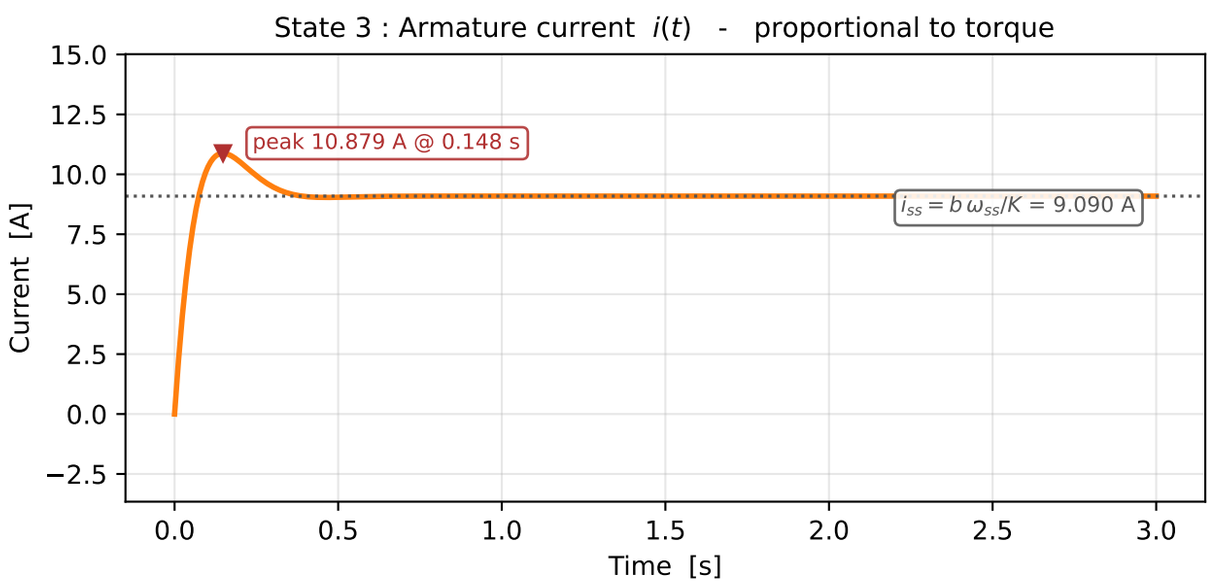
PD controller Kp = 100.00 Ki = 0.00 Kd = 5.00 Tf = 0.000 s
motor: J = 0.0100 b = 0.1000 K = 0.0100 R = 1.000 L = 0.500 H derivative on measurement

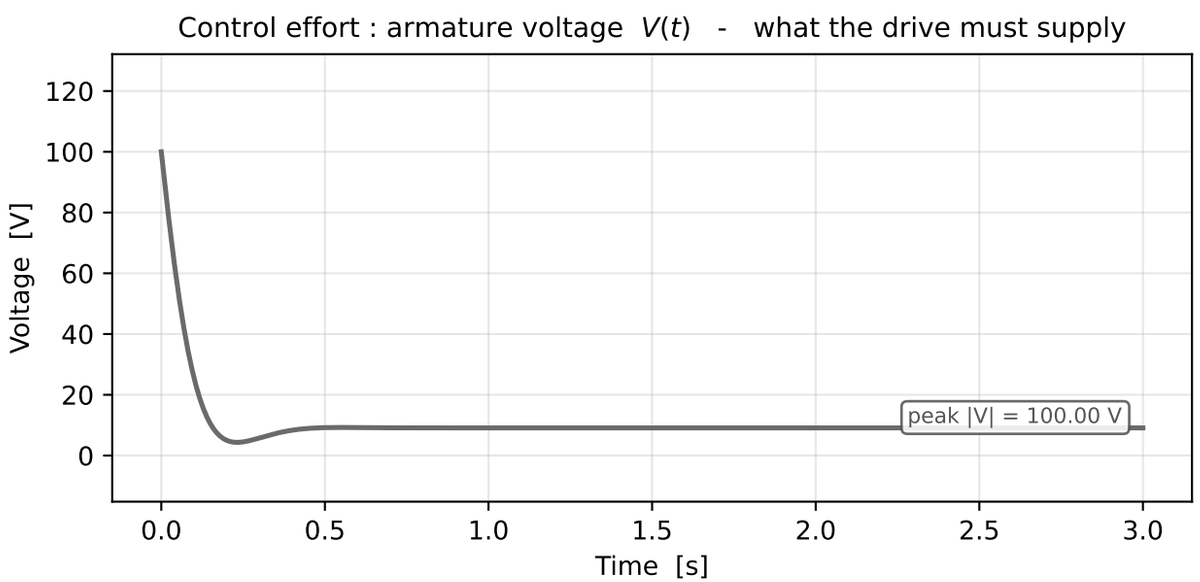
STABLE - every closed-loop pole lies in the left half-plane (rightmost Re = -11.000)

`closed-loop poles:  -11.000+9.951j, -11.000-9.951j`





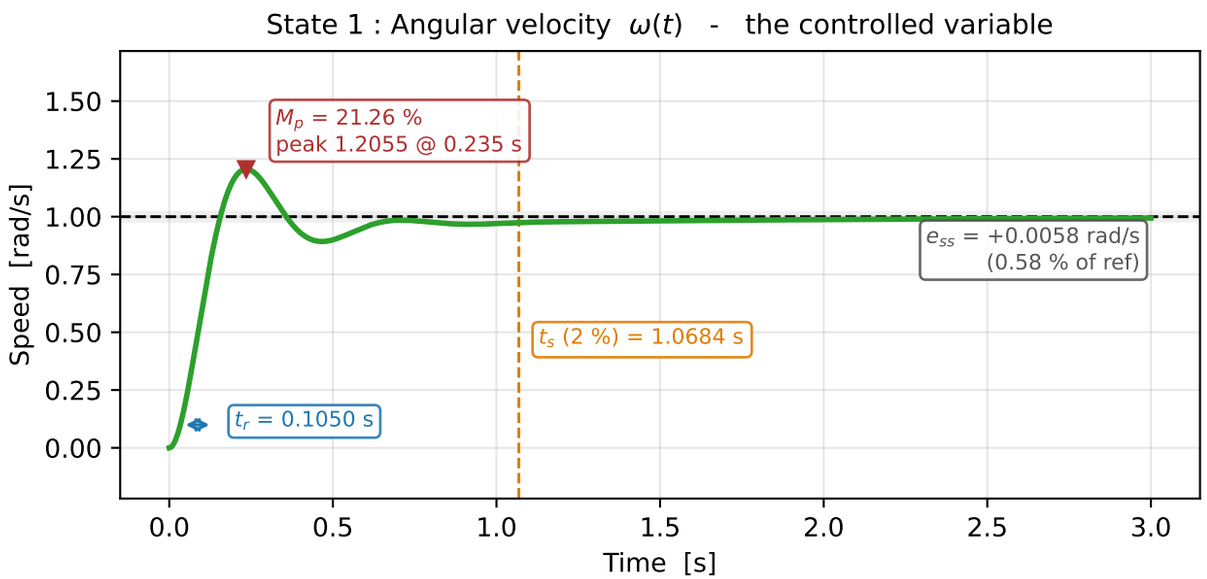


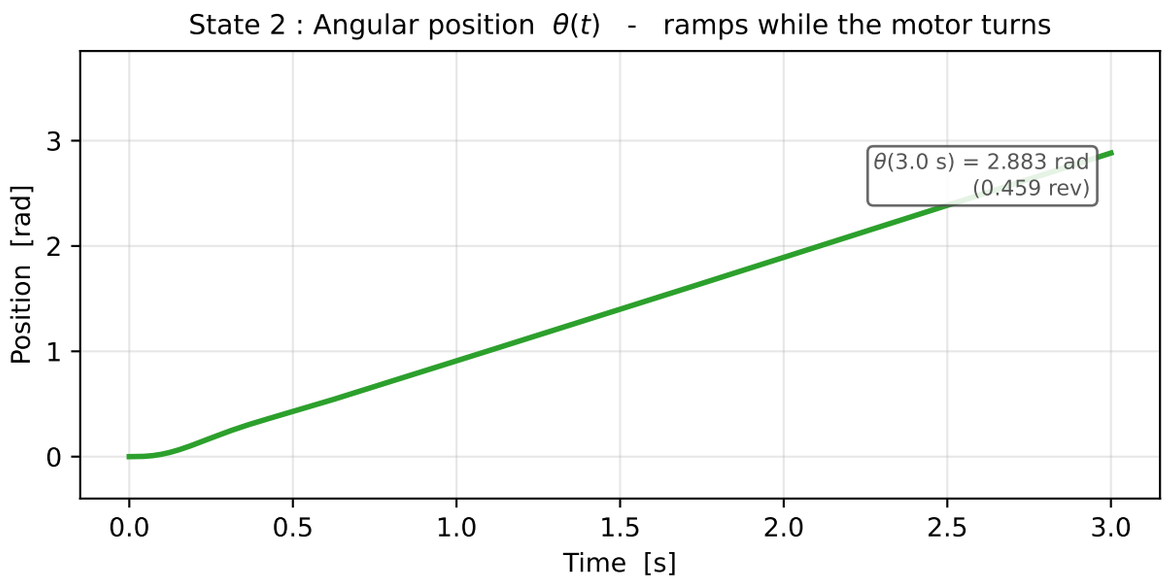


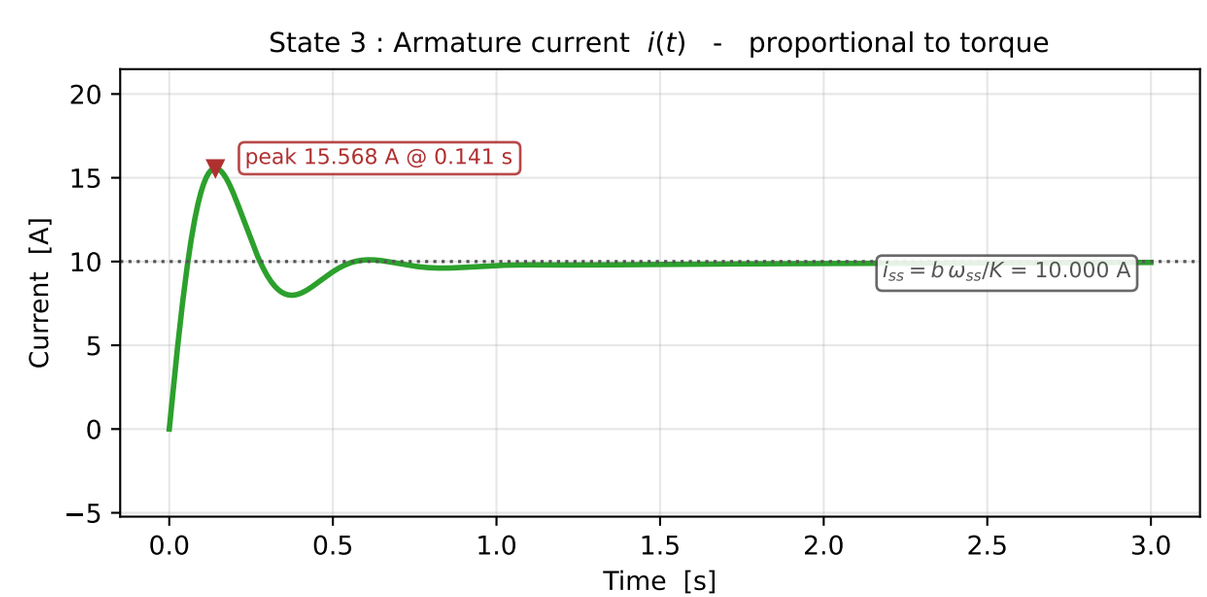
PI controller Kp = 100.00 Ki = 80.00 Kd = 0.00 Tf = 0.000 s
motor: J = 0.0100 b = 0.1000 K = 0.0100 R = 1.000 L = 0.500 H derivative on measurement

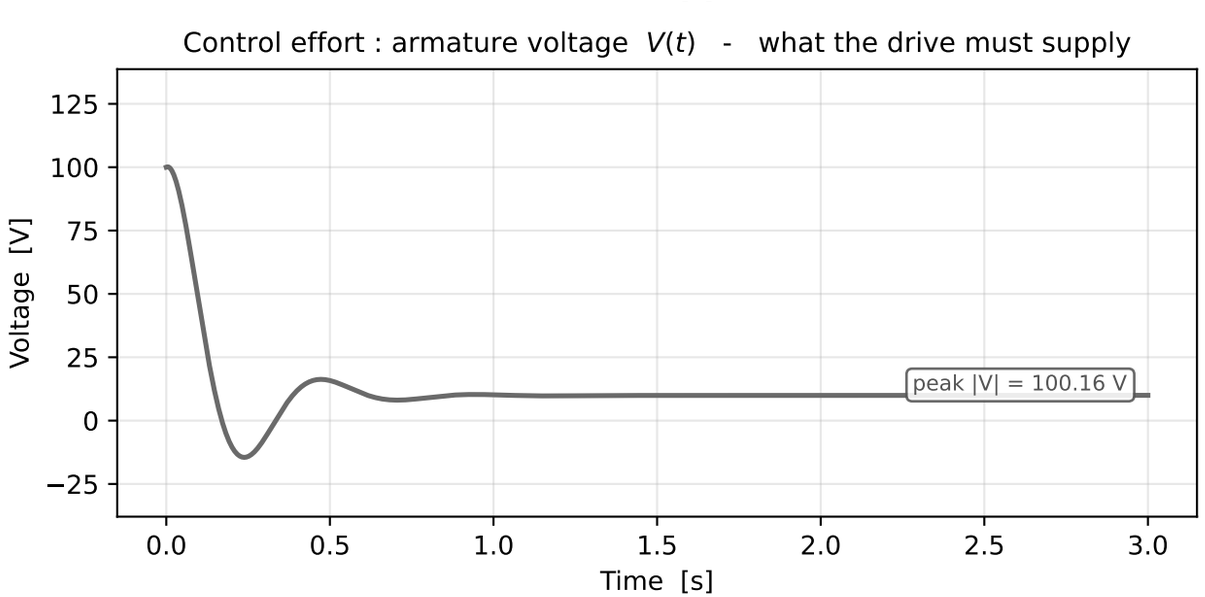
STABLE - every closed-loop pole lies in the left half-plane (rightmost Re = -0.756)

closed-loop poles: -5.622+13.413j, -5.622-13.413j, -0.756







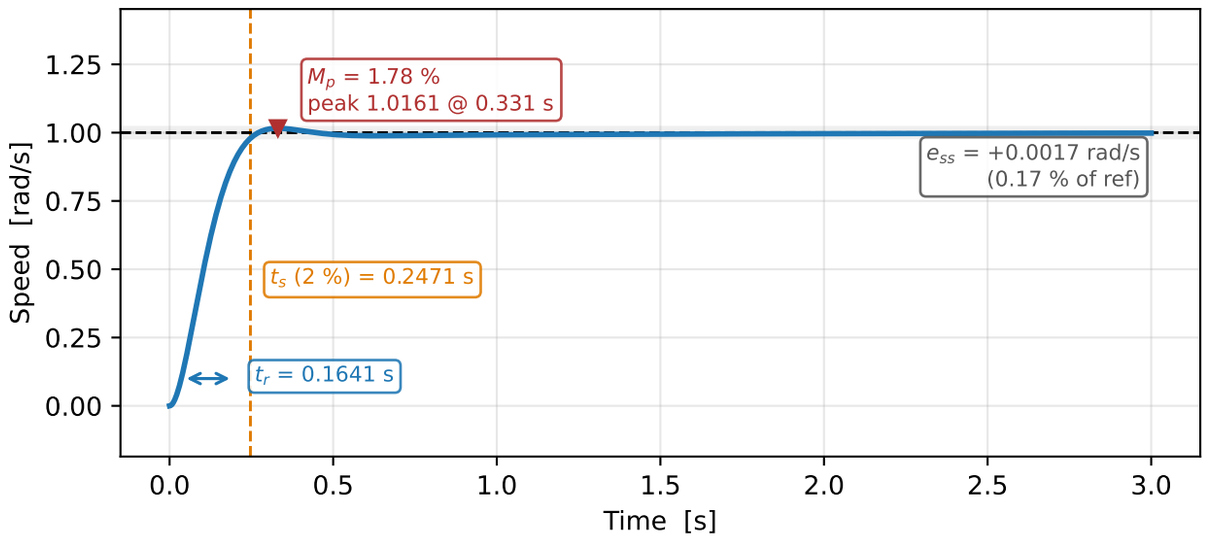


PID controller Kp = 100.00 Ki = 80.00 Kd = 5.00 Tf = 0.000 s
motor: J = 0.0100 b = 0.1000 K = 0.0100 R = 1.000 L = 0.500 H derivative on measurement

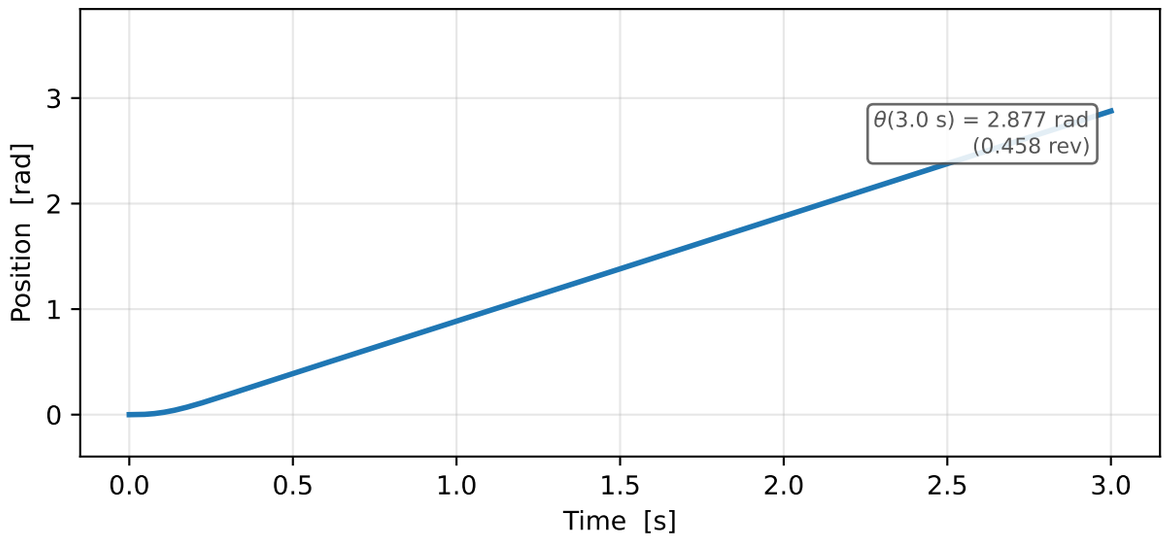
STABLE - every closed-loop pole lies in the left half-plane (rightmost Re = -0.787)

closed-loop poles: -10.607+9.530j, -10.607-9.530j, -0.787

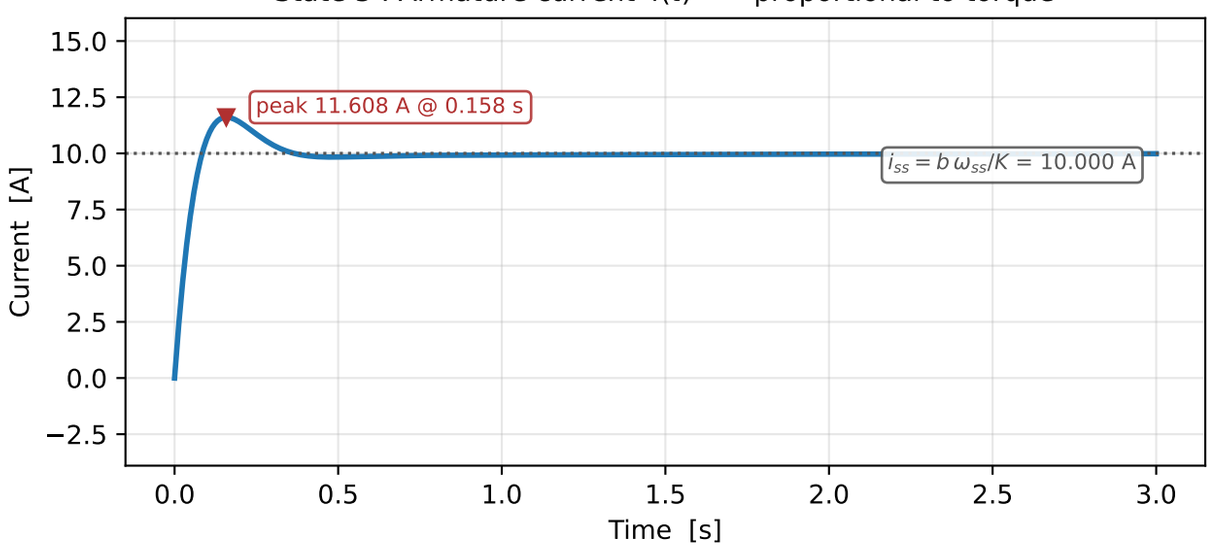
State 1 : Angular velocity $\omega(t)$ - the controlled variable



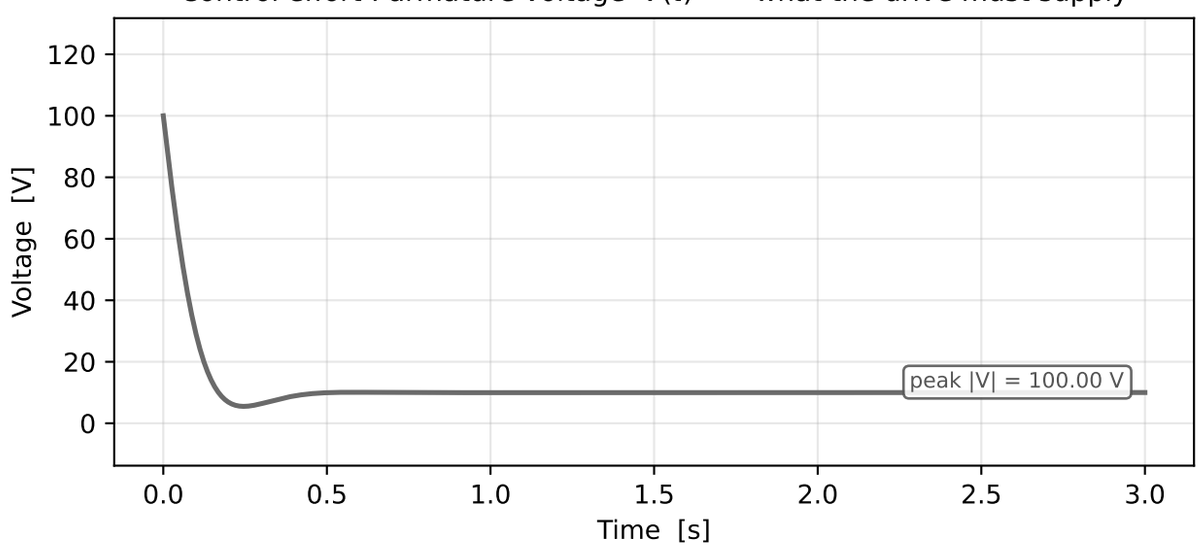
State 2 : Angular position $\theta(t)$ - ramps while the motor turns



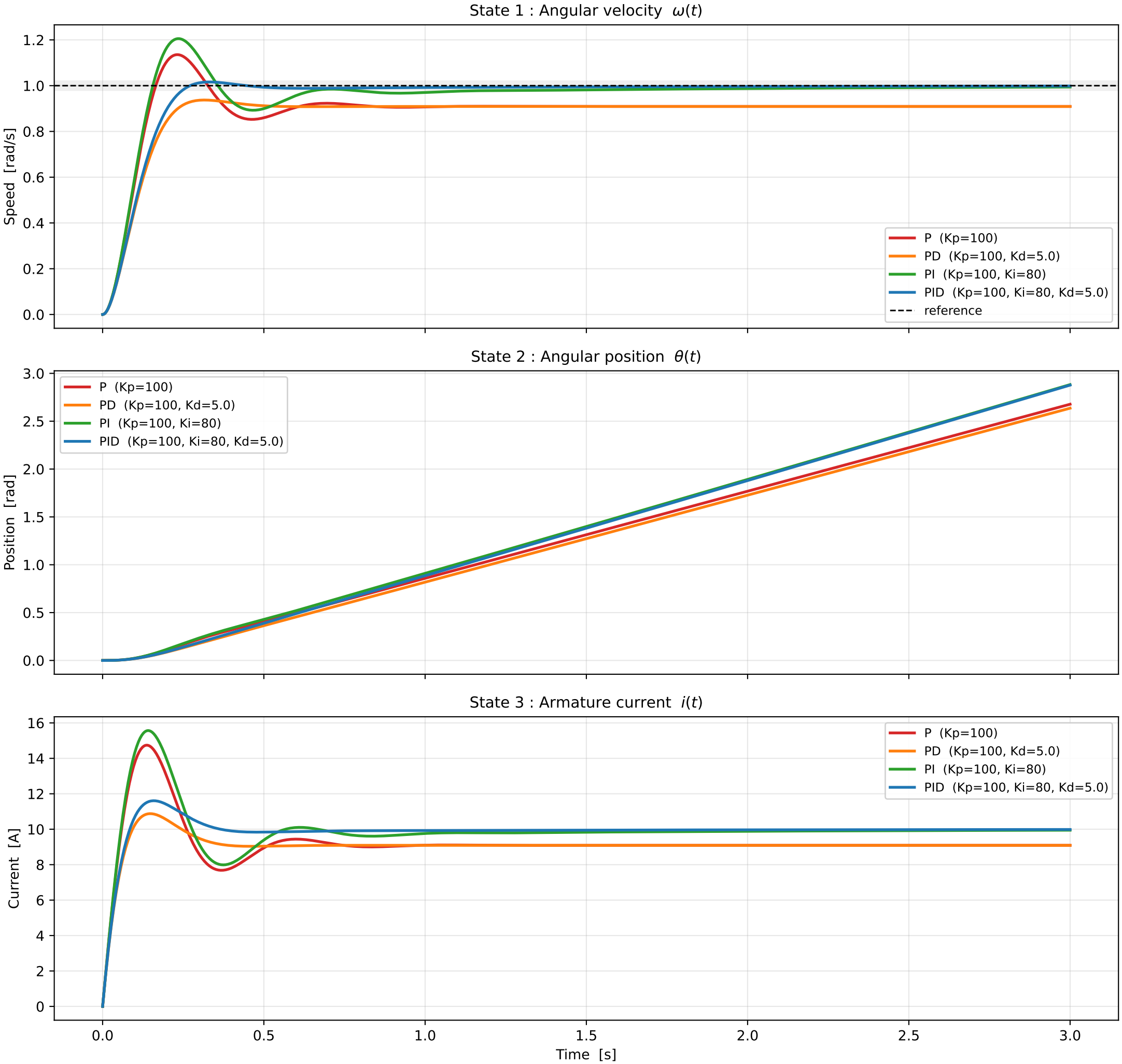
State 3 : Armature current $i(t)$ - proportional to torque



Control effort : armature voltage $V(t)$ - what the drive must supply



P vs PD vs PI vs PID Kp = 100.00 Ki = 80.00 Kd = 5.00 Tf = 0.000 s
motor: J = 0.0100 b = 0.1000 K = 0.0100 R = 1.000 L = 0.500 H derivative on measurement



PERFORMANCE METRICS - speed loop

step reference = 1.00 rad/s rise time 10-90 % settling band +/-2 % overshoot relative to the final value

Controller	Kp	Ki	Kd	tr (s)	ts (s)	Mp (%)	ess (rad/s)	peak i (A)	peak V (V)	Stability
P	100.0	0.0	0.00	0.0990	0.5672	24.92	0.09099	14.743	100.00	STABLE
PD	100.0	0.0	5.00	0.1521	0.3921	3.10	0.09099	10.879	100.00	STABLE
PI	100.0	80.0	0.00	0.1050	1.0684	21.26	0.00000	15.568	100.16	STABLE
PID	100.0	80.0	5.00	0.1641	0.2471	1.78	0.00000	11.608	100.00	STABLE

MOST SUITABLE CONTROLLER : PID

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Zero steady-state error is reached only by : PI, PID
  - a derivative term cannot help here, since the derivative of a
    constant is zero, so it contributes nothing at DC;
  - only the integrator raises the loop to type 1.

Among those, the shortest settling time belongs to PID :
  tr = 0.1641 s     ts = 0.2471 s     Mp = 1.78 %
  peak current 11.608 A     peak voltage 100.00 V

P   is quickest to rise but overshoots 24.9 % and keeps a 0.0910 rad/s offset.
PD  damps that overshoot to 3.1 % yet the offset is untouched.
PI  removes the offset but is the slowest to settle at 1.0684 s.
PID is the only structure that is both on target and well damped.
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